

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

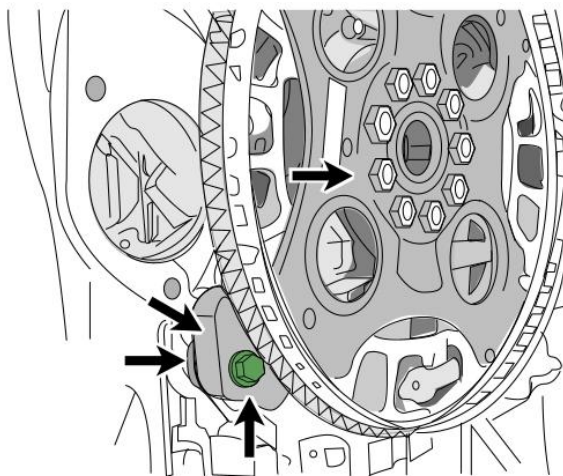
Always use new gaskets

Removal

Remove the engine

Vehicles with automatic transmission:

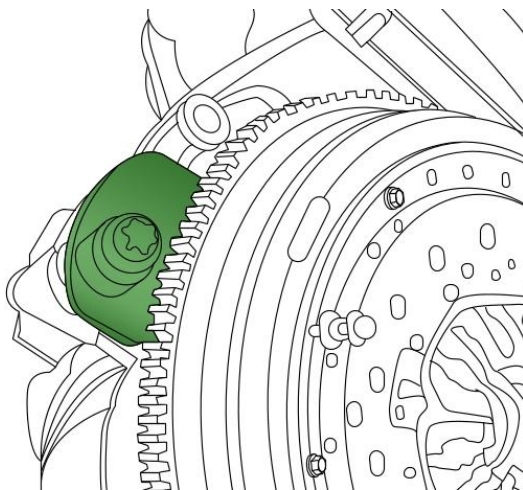
Fit the locking tool (11 9 260)



Remove the flexplate

Vehicles with manual gearbox:

Fit the flywheel locking tool (11 9 810, 11 9 811)



Remove the flywheel

All vehicles:

Remove the cylinder head

Remove the oil sump

Remove the oil pump

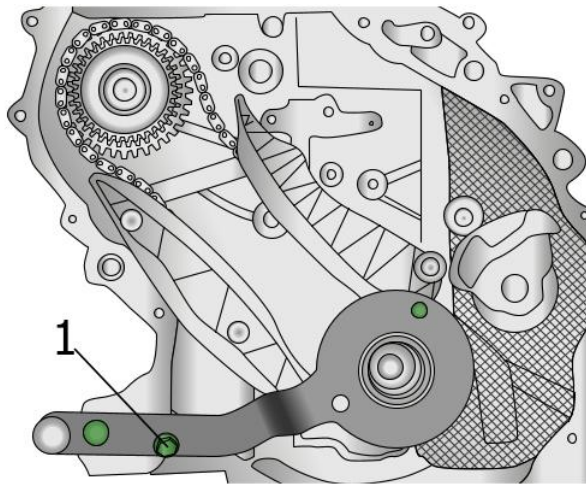
Vehicles with automatic transmission:

Repair Manuals

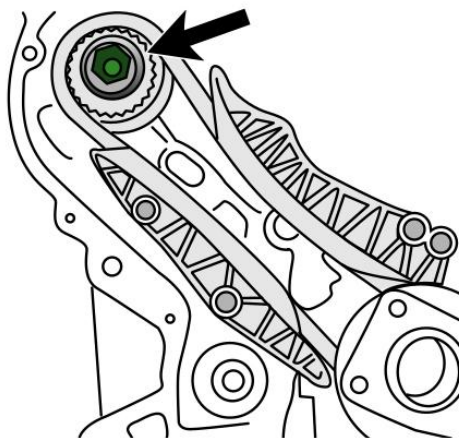
Remove the oil cooler
Remove the oil cooler bracket

All vehicles:

Remove the crankshaft position sensor ring
Remove the timing chain cover
Remove the oil pump chain
Remove the primary timing chain
Remove the upper timing chain guide
Fit the crankshaft locking tool (2 357 904)
Fit the special tool at the point shown
Fit the bolt (1)

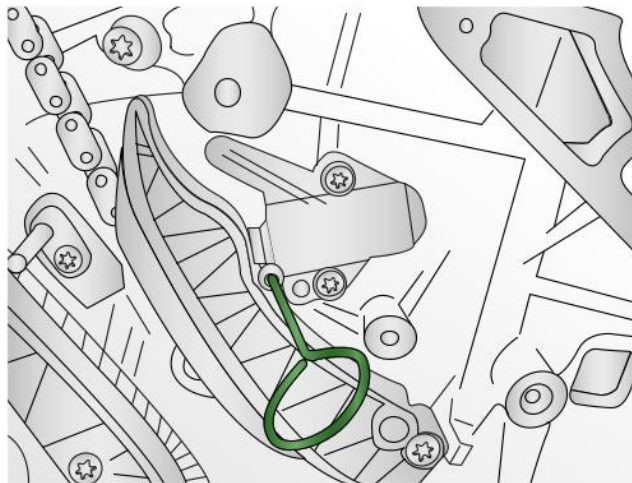


Loosen the high-pressure pump gearwheel bolt
Fit the special tool: (11 8 740)
Detach the high-pressure pump gearwheel



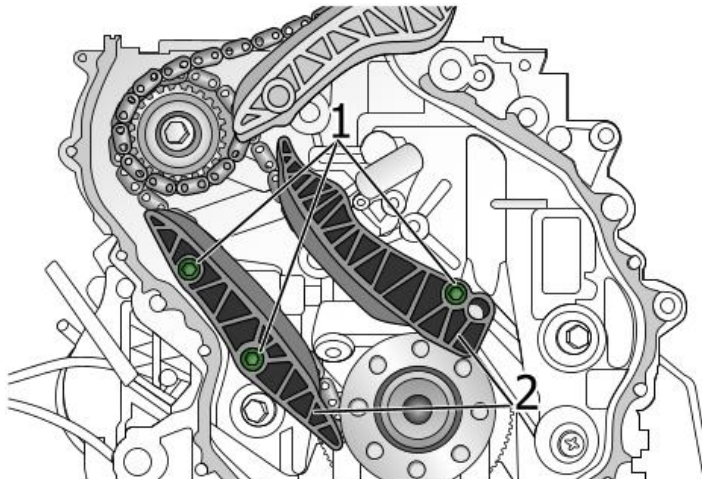
Lock the chain tensioner with a locking pin (11 4 120)

Repair Manuals



Remove the secondary chain tensioner

Remove the timing chain guides (1 - 2)



Remove the high-pressure pump gearwheel

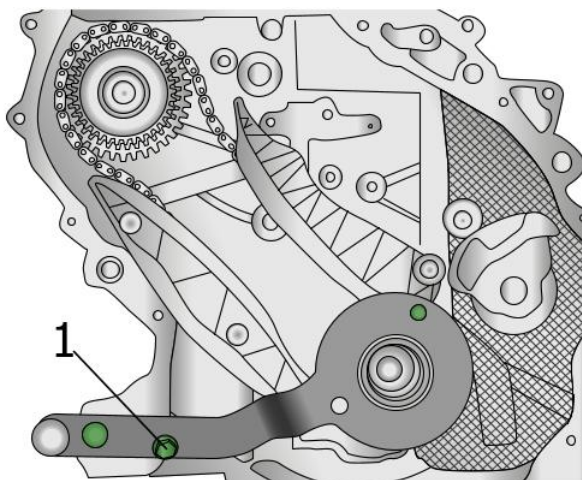
Remove the crankshaft locking tool (2 357 904)

Remove the secondary timing chain

Installation

Refit the secondary timing chain

Fit the crankshaft locking tool

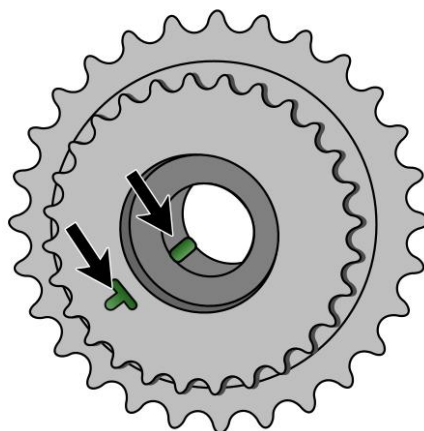
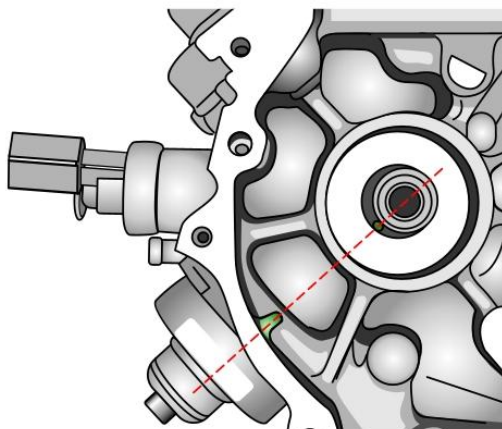


Repair Manuals

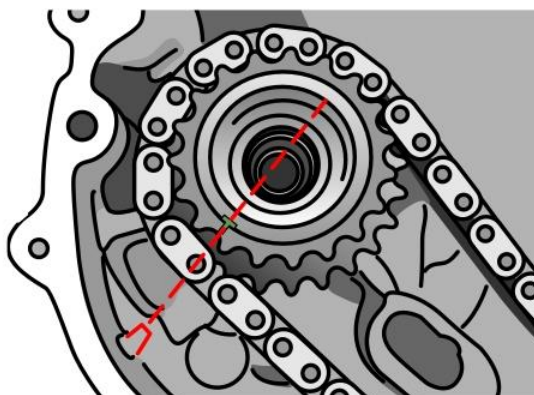
Fit the special tool at the point shown

Fit the bolt (1)

Check the timing marks



Refit the high-pressure pump gearwheel



Timing chain cover:

Check the dowel pin(s) for damage

Apply sealant

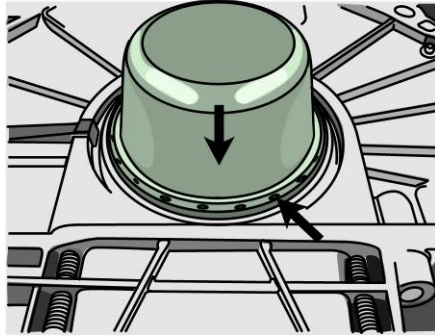
Repair Manuals

Refit the timing cover

Crankshaft oil seal:

Use the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Fit the special tool: (11 8 815)



Note: Do not apply grease

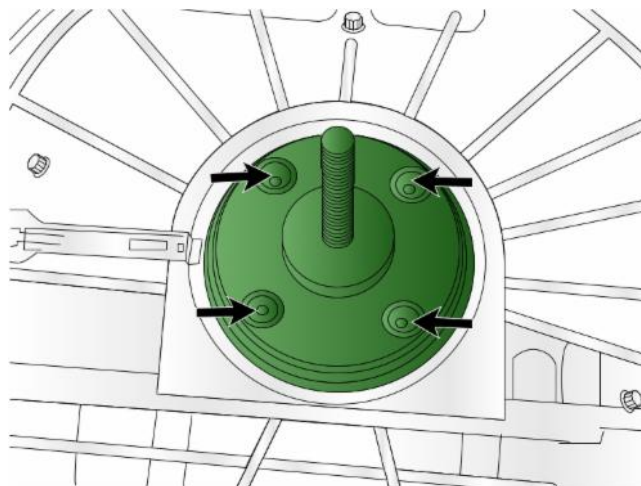
Fit the crankshaft seal over the special tool

Note: Do not touch the lip of the seal

Remove the special tool (11 8 815)

Fit the special tool: (11 8 811)

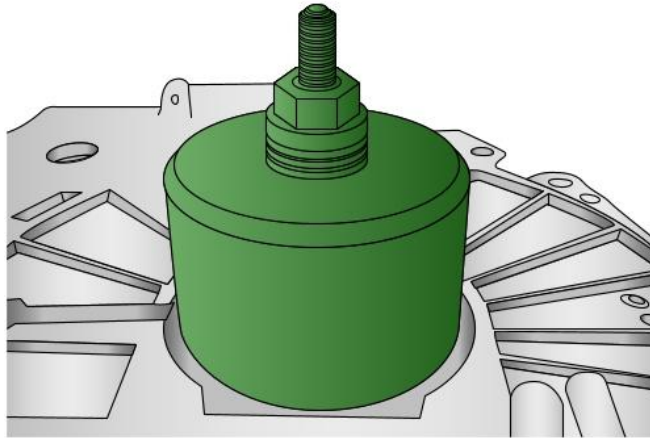
Tighten the bolts (11 8 812)



Fit the special tool: (11 8 813)

Tighten the nut (11 8 814)

Repair Manuals



Note: Until the crankshaft seal is level with the timing cover

Remove the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Further refitting is in the reverse order of removal

Torque settings

Oil sump

Sump 24 (Nm)

Oil pump

Oil pump 22 (Nm)

Oil pump gearwheel

Oil pump gearwheel (Renew the bolt (left-hand thread))

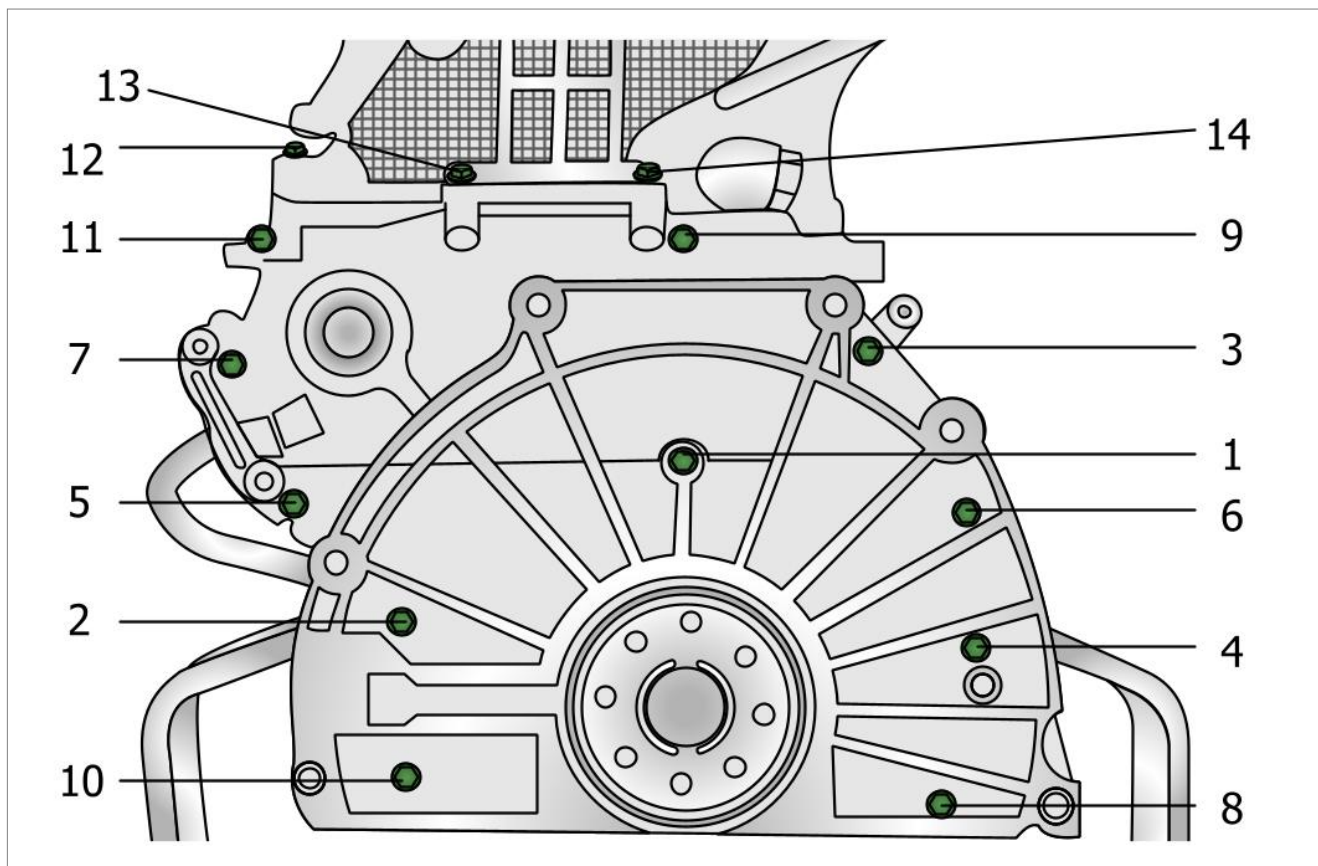
Stage 1 5 (Nm)

Stage 2 90 (°)

Timing cover

Timing cover (Renew the bolts)

Repair Manuals



Timing cover (Bolts 1 - 11)

Stage 1 8 (Nm)

Stage 2 90 (°)

Timing cover (Bolts 12 - 14) 19 (Nm)

Timing chain tensioner

Timing chain tensioner (Upper) 70 (Nm)

Timing chain tensioner (Lower) 10 (Nm)

Timing chain guide

Timing chain guide (Renew the bolts) 20 (Nm)

High-pressure pump gearwheel

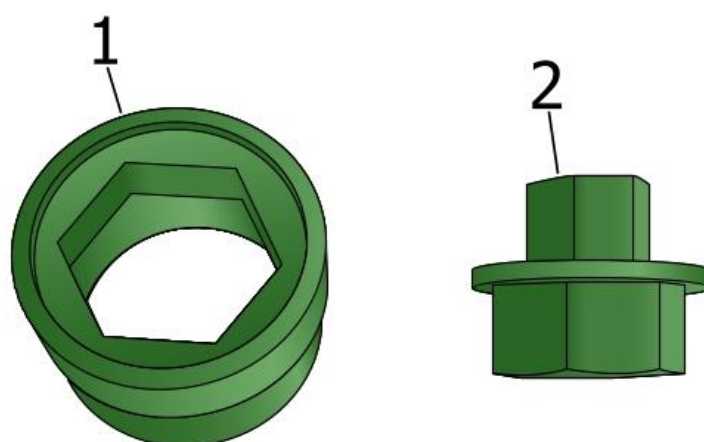
High-pressure pump gearwheel 65 (Nm)

Flywheel

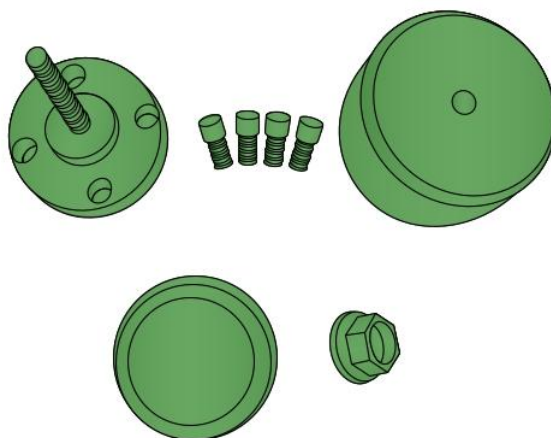
Flywheel (Renew the bolts) 120 (Nm)

Special tools

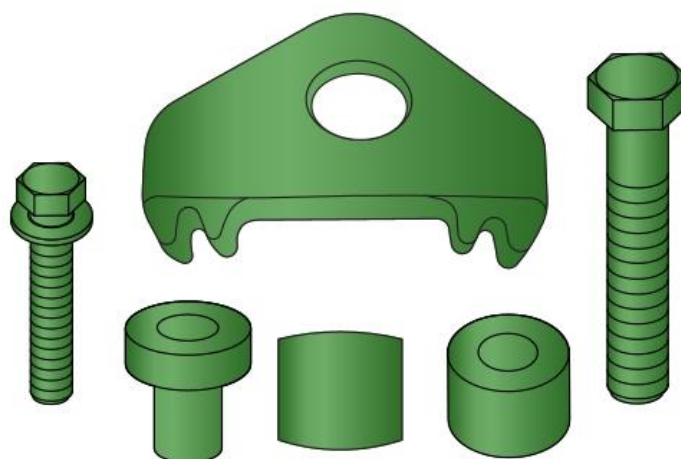
Extractor: (11 8 740)

Repair Manuals

Oil seal fitting tool: (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

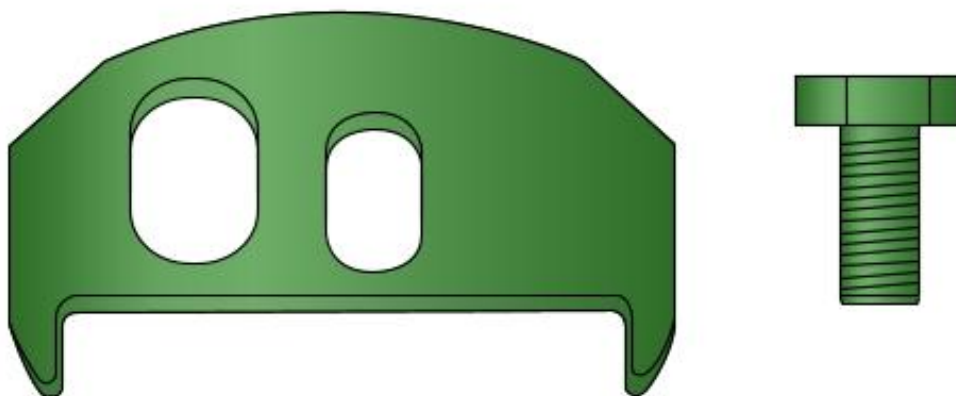


Flywheel locking tool: (11 9 260)

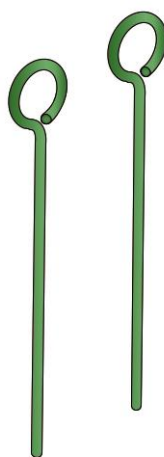


Flywheel locking tool: (11 9 811)

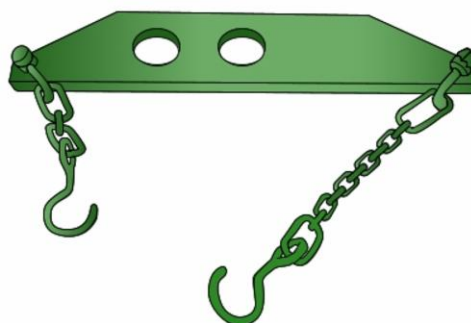
Repair Manuals



Locking pin: (11 4 120)

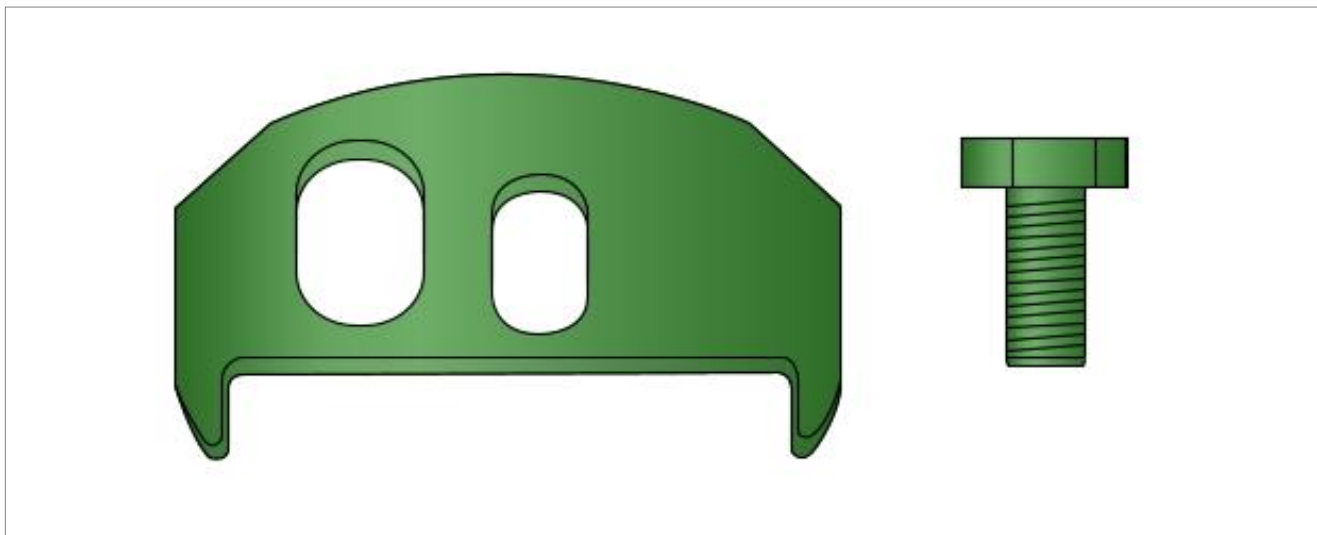


Engine support tool: (11 0 000)



Flywheel locking tool: (11 9 810)

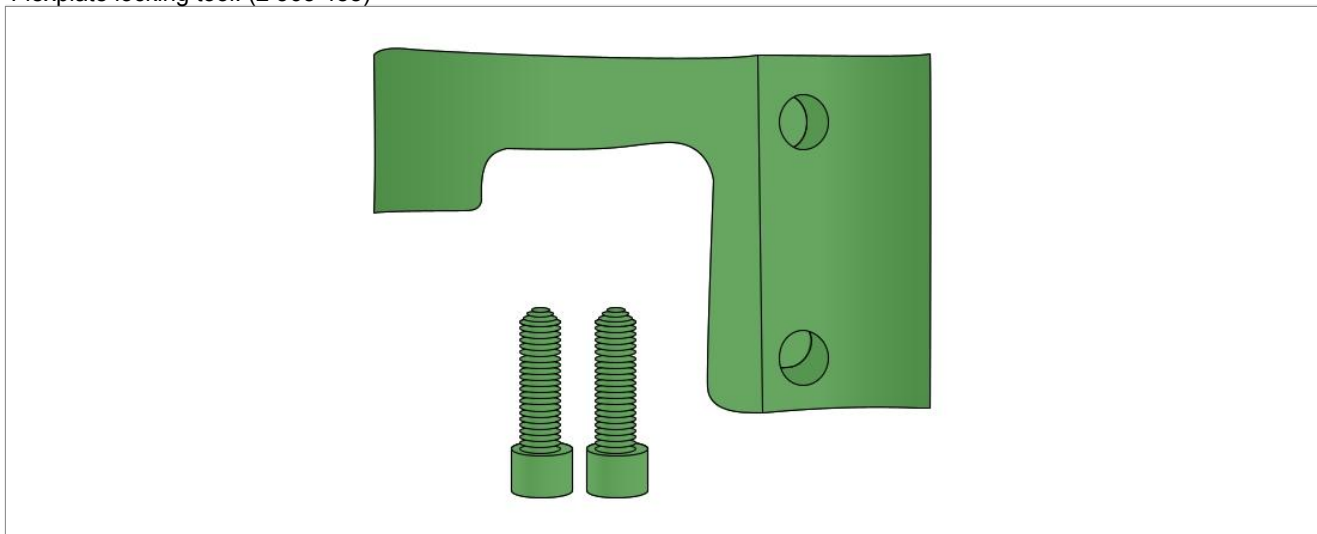
Repair Manuals



Crankshaft locking tool: (2 357 904)



Flexplate locking tool: (2 365 488)



Flexplate locking tool: (2 288 380)

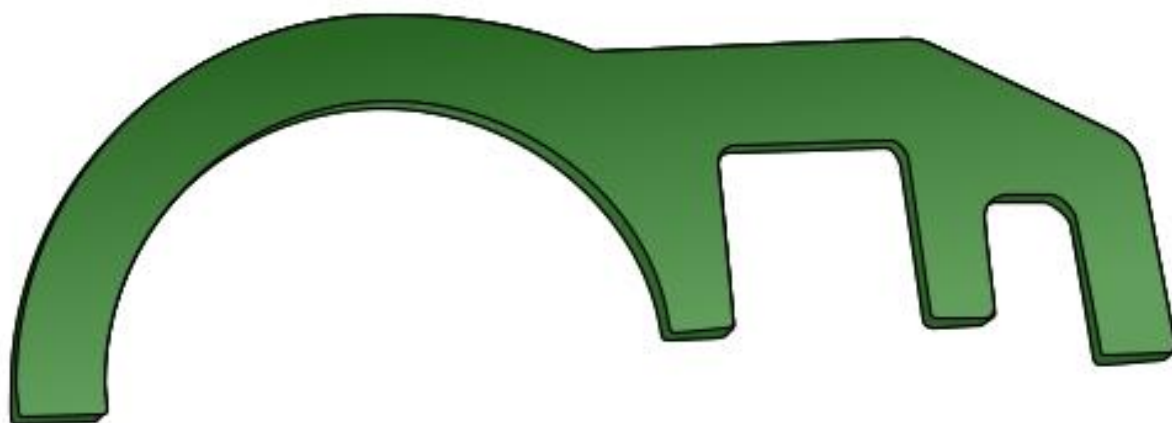
Repair Manuals



Socket: (11 6 480)

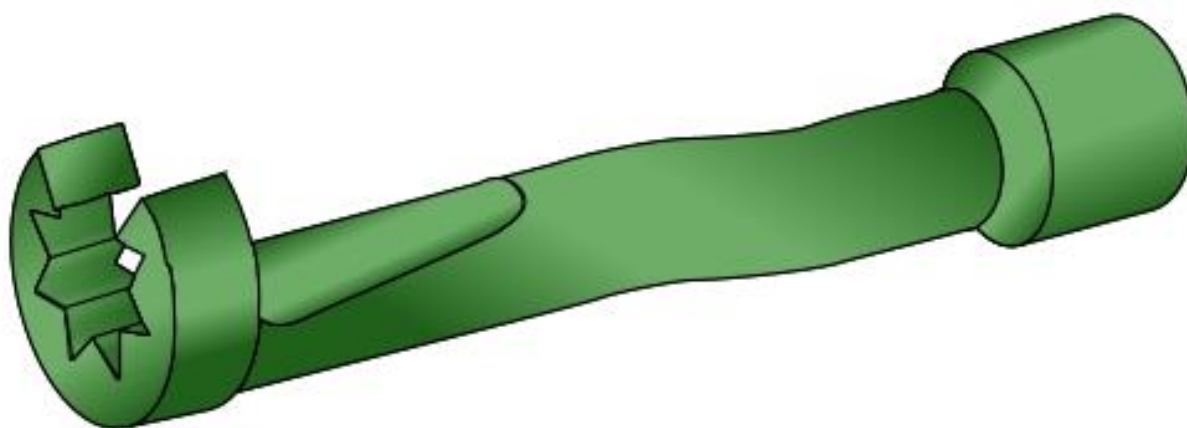


Camshaft blocking tool: (11 8 760)

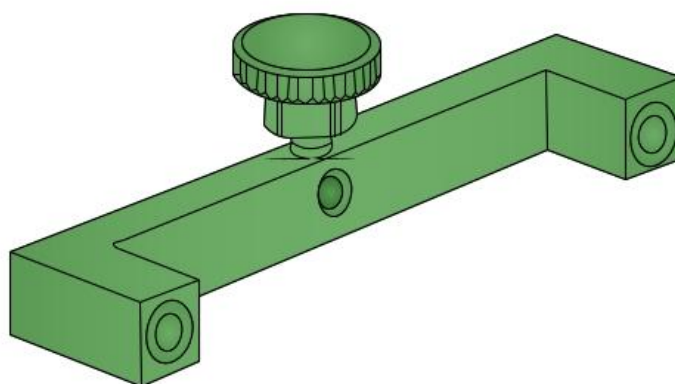


Socket wrench: (37 1 152)

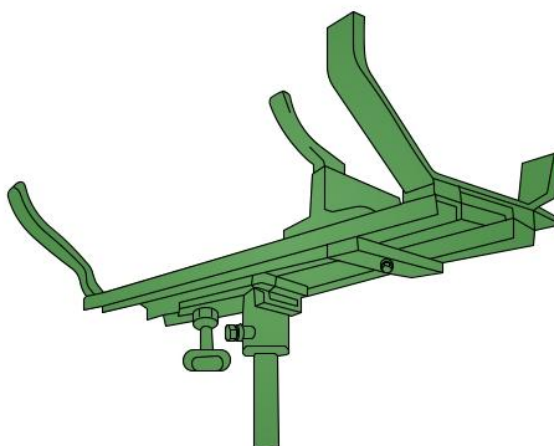
Repair Manuals



Dial gauge holder: (00 2 530)

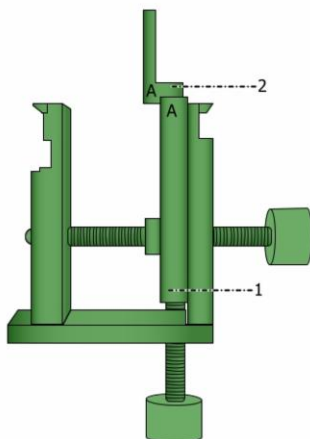


Gearbox support: (23 4 050, 00 2 030)

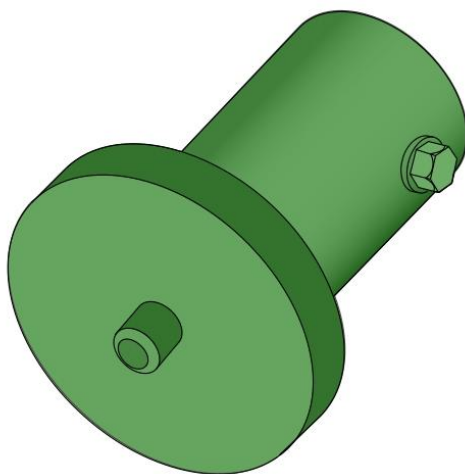


Bearing bracket: (24 4 160, 24 4 161, 24 4 164)

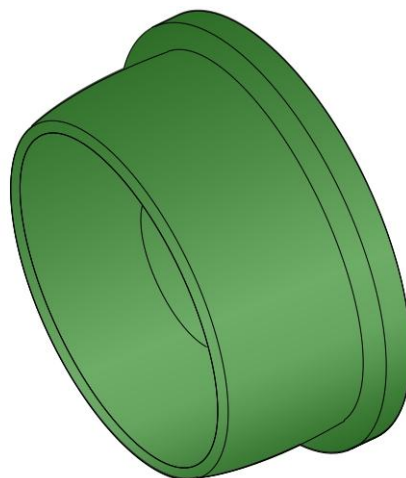
Repair Manuals



Support (31 2 220)

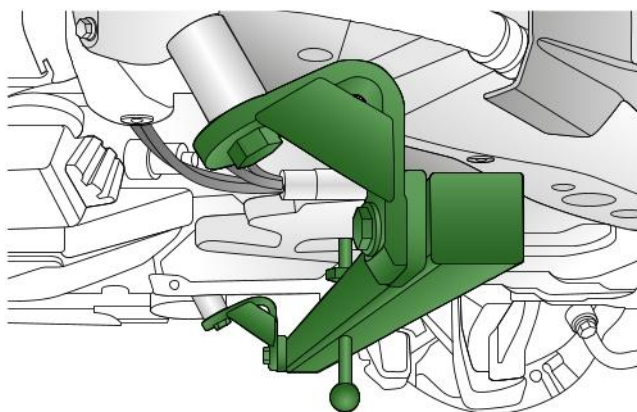


Plug (24 2 390)

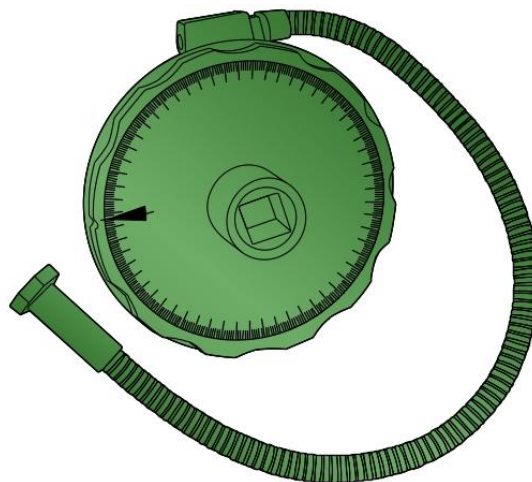


Gearbox support tool: (11 8 910)

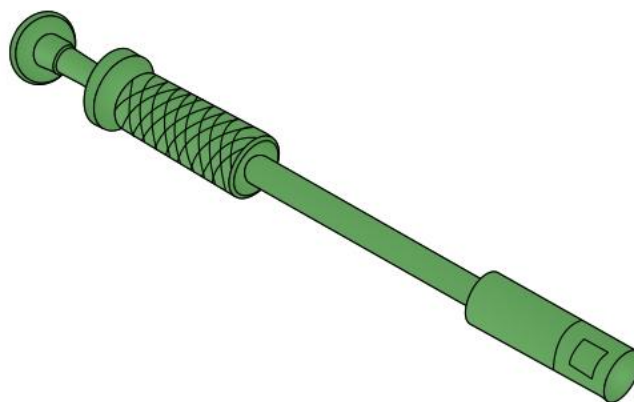
Repair Manuals



Torque wrench: (00 9 120)



Injector removal tool: (13 5 250)

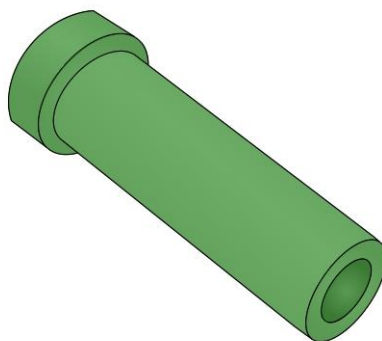


Injector removal tool: (13 5 252)

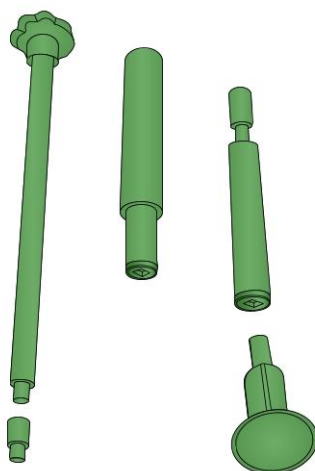
Repair Manuals



Injector removal tool: (13 5 253)

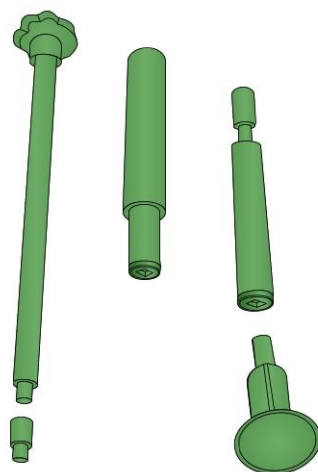


Injector cleaner: (13 5 215)

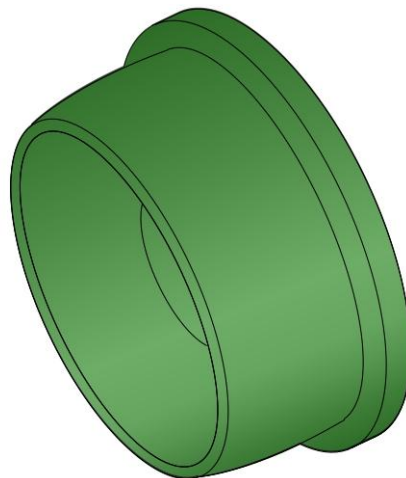


Injector cleaner: (13 5 210)

Repair Manuals



Plug (24 3 390)

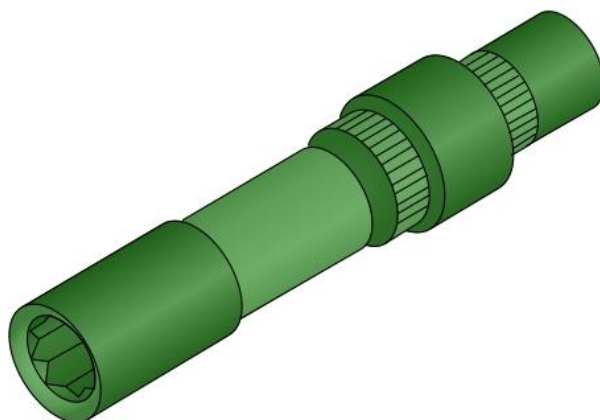


Impact extractor: (13 6 100)

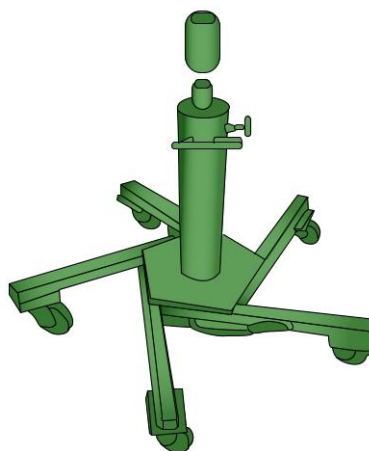


Socket: (11 6 050)

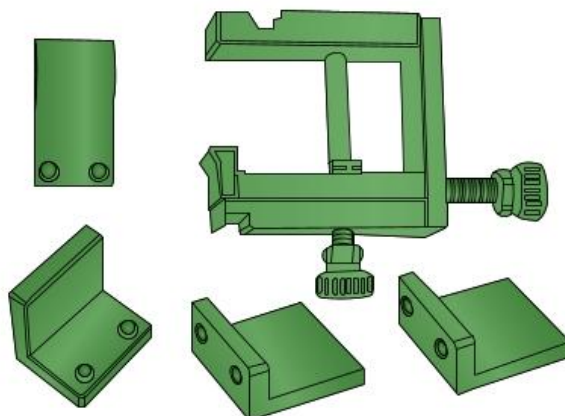
Repair Manuals



Transmission jack: (00 2 030)



Locking tool: (24 4 161, 24 4 165)



Gearbox support tool: (23 4 050)

Repair Manuals

